

Synoptic CB: Macrophyte Fluxes

2025 Samples

2026-03-16

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#Read in and collate all the files in the Processed Data folder

##Pull in Hydrago probe data

Some flux IDs are missing from hydrago data

```
## [1] "GCW_TR_1_202509" "GCW_TR_2_202509" "GCW_TR_3_202509" "GCW_TR_4_202509"
## [5] "GCW_TR_5_202509" "GCW_WC_1_202509" "GCW_WC_2_202509" "GCW_WC_3_202509"
## [9] "GCW_WC_4_202509" "GCW_WC_5_202509" "GWI_WC_1_202503" "GWI_WC_1_202507"
## [13] "GWI_WC_2_202503" "GWI_WC_2_202507" "GWI_WC_3_202503" "GWI_WC_3_202507"
## [17] "GWI_WC_4_202503" "GWI_WC_4_202507" "GWI_WC_5_202503" "GWI_WC_5_202507"
## [21] "MSM_TR_1_202507" "MSM_TR_2_202507" "MSM_TR_3_202507" "MSM_TR_4_202507"
## [25] "MSM_TR_5_202507" "MSM_WC_1_202507" "MSM_WC_2_202507" "MSM_WC_3_202507"
## [29] "MSM_WC_4_202507" "MSM_WC_5_202507" "SWH_WC_1_202505" "SWH_WC_1_202509"
## [33] "SWH_WC_2_202505" "SWH_WC_2_202509" "SWH_WC_3_202505" "SWH_WC_3_202509"
## [37] "SWH_WC_4_202505" "SWH_WC_4_202509" "SWH_WC_5_202505" "SWH_WC_5_202509"
```

##Remove bad fluxes and calculate how many there were

Number flux measurements removed: 15

Number negative CO2 fluxes removed: 1

Number unusually high starting value fluxes removed: 3

Number messy fluxes removed: 11

Percentage flux measurements removed: 8.82%

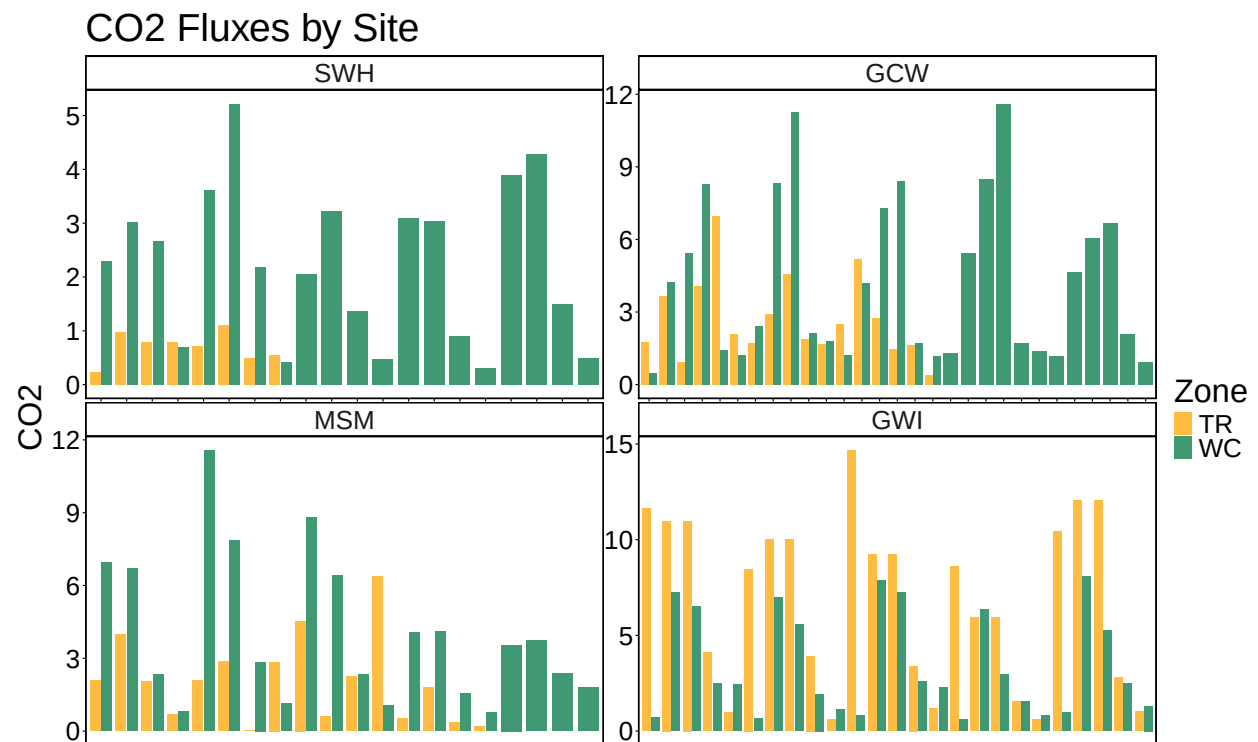
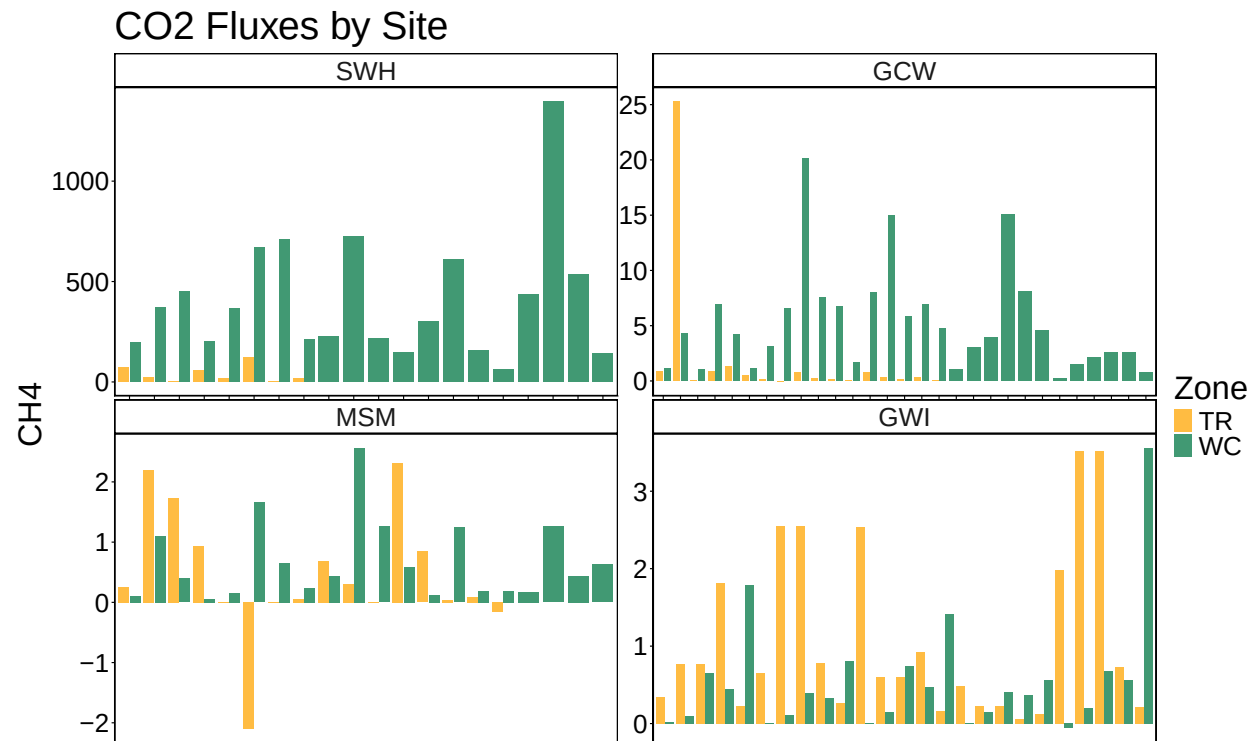
Percentage negative CO2 fluxes removed: 0.59%

Percentage unusually high starting value fluxes removed: 1.76%

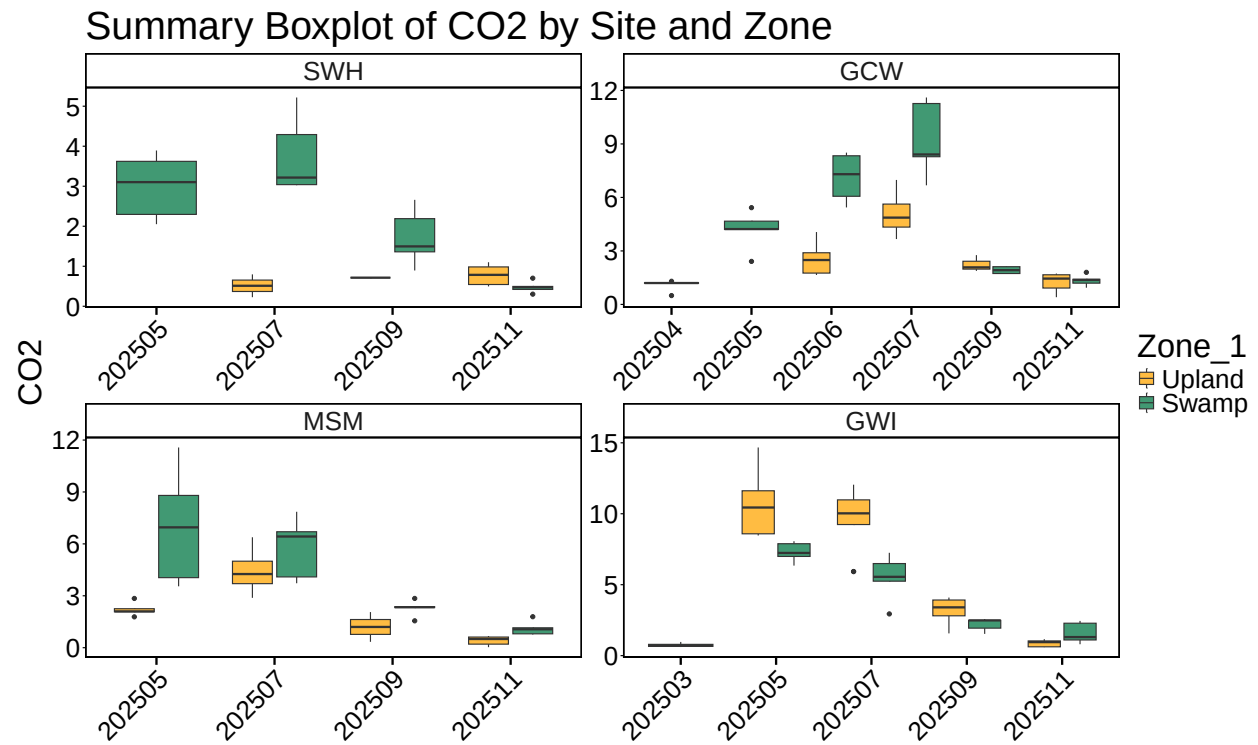
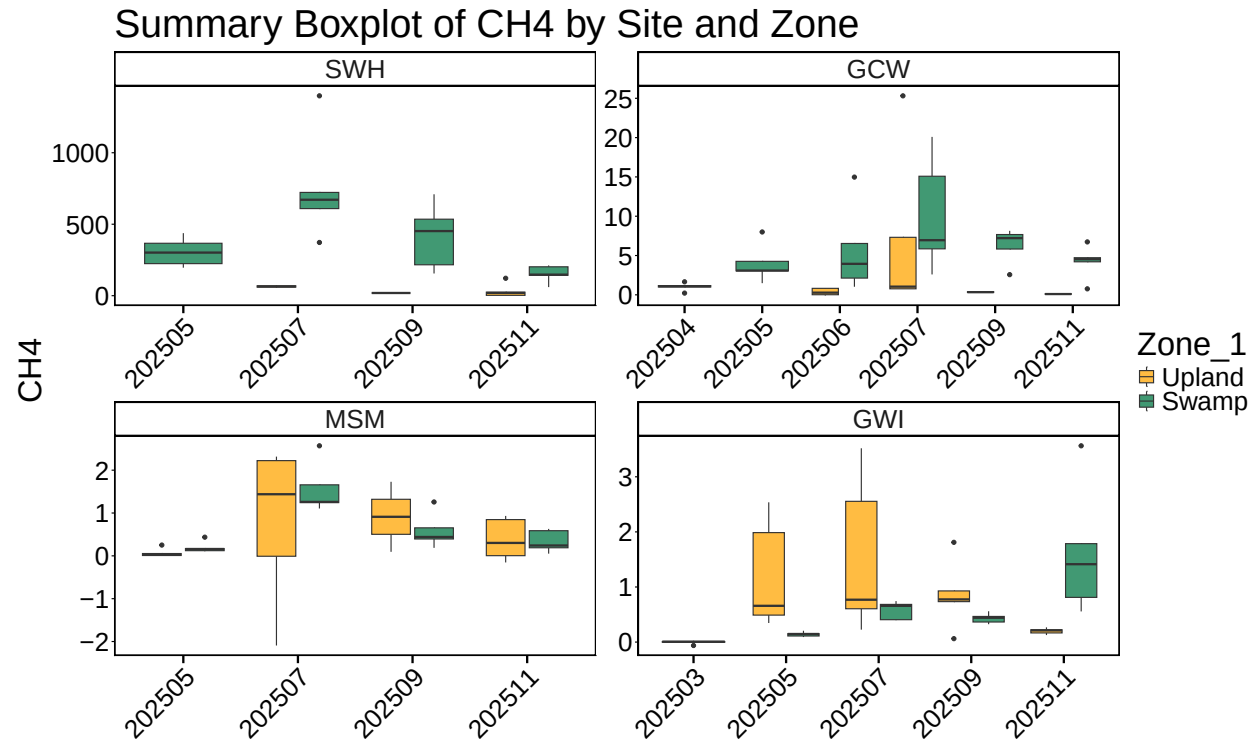
Percentage messy fluxes removed: 6.47%

0.1 Write out files

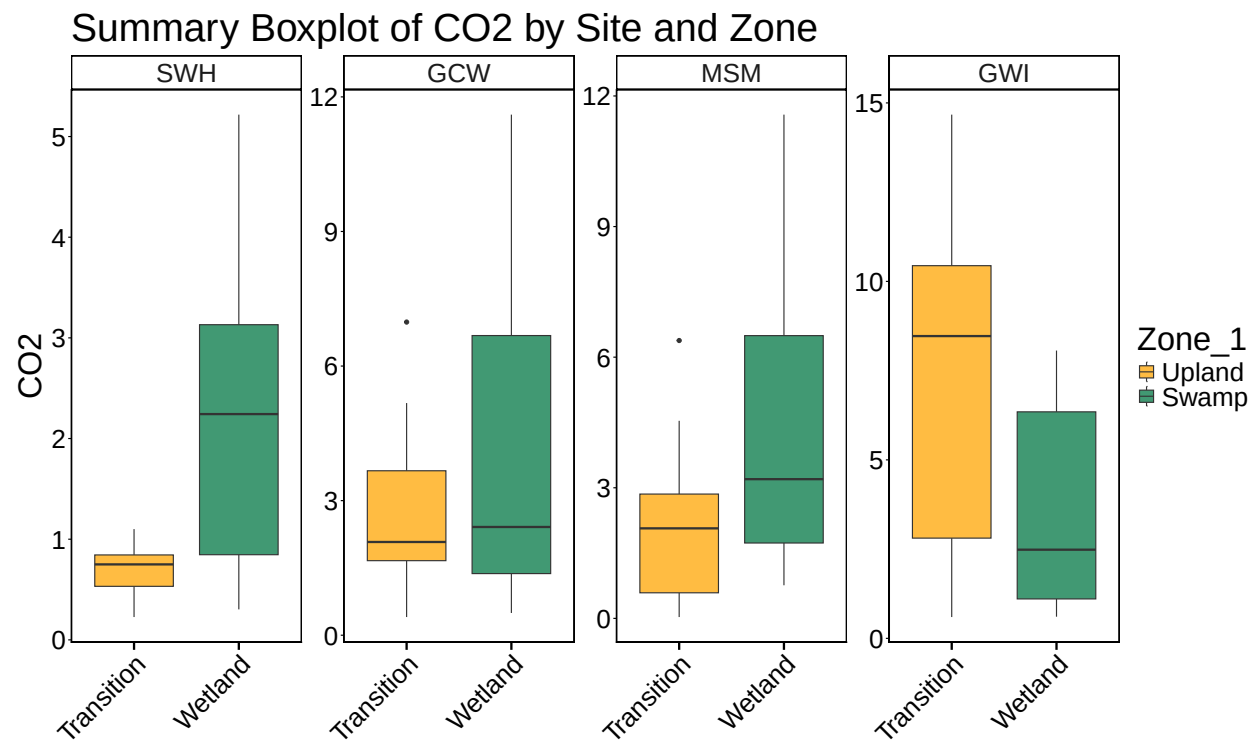
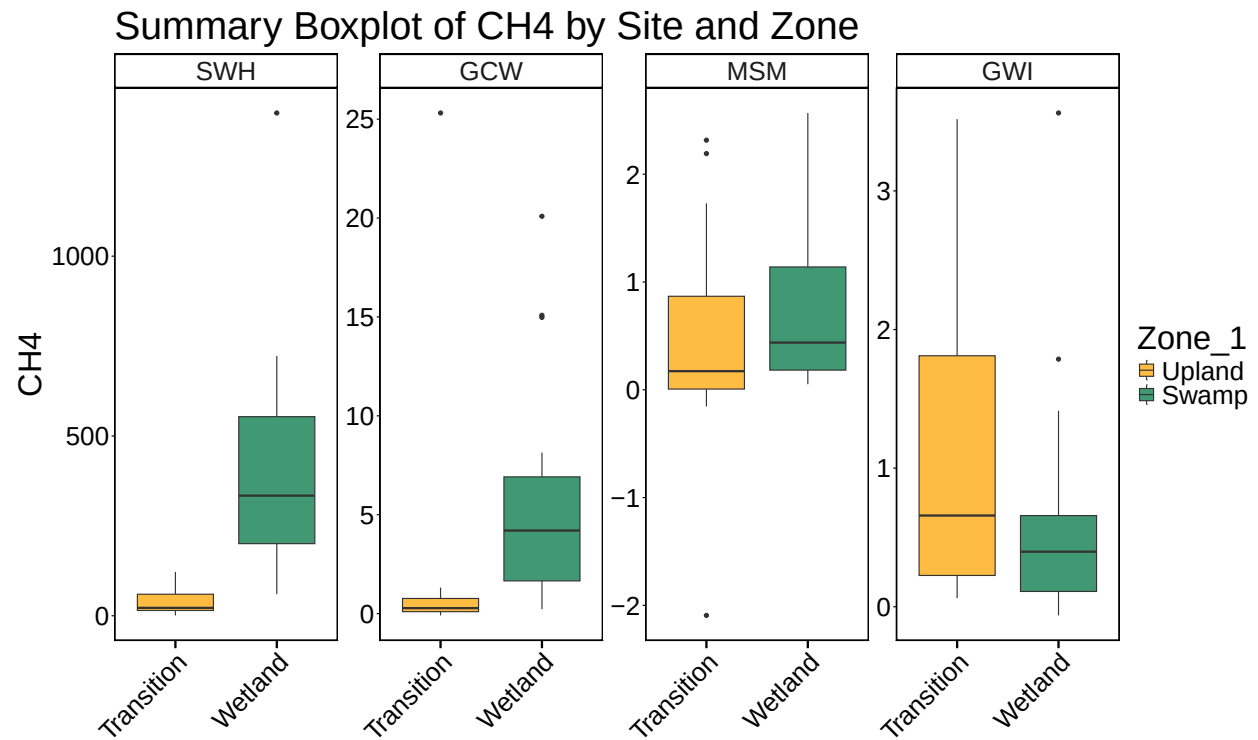
0.2 All Data by Year, Site and Zone



0.3 Boxplot summary by Date



0.4 Boxplot summary by Site



##Calculate summary metrics

0.5 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
SWH	Transition	1.0872692	121.977500	39.7087775	0.2266990	1.100404	0.7058451
SWH	Wetland	59.9871778	1398.120534	406.5684788	0.3028537	5.217695	2.2376064
GCW	Transition	-0.0901019	25.312906	1.8577613	0.4050037	6.980726	2.7131738
GCW	Wetland	0.2215926	20.090639	5.1816997	0.4957703	11.604588	4.2294688
MSM	Transition	-2.0931613	2.316486	0.4504229	0.0305177	6.382414	2.0786885
MSM	Wetland	0.0514141	2.568636	0.6728730	0.7602931	11.571173	4.0361589
GWI	Transition	0.0621000	3.517843	1.0666977	0.5968420	14.671811	6.8158948
GWI	Wetland	-0.0623885	3.562404	0.5542746	0.6065226	8.062884	3.4680774